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Advancing Wellbore Stability: A Data-Driven Study on Lost Circulation Spacer Efficiency Application in the Middle East

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Abstract

Lost circulation is a critical challenge in Middle Eastern oil and gas operations. This paper analyzes six years of data on the efficiency of a lost circulation spacer, highlighting its role in reducing fluid losses during cementing operations. The study identifies key success factors and failure conditions, offering insights to improve treatment outcomes.

The lost circulation spacer is an engineered fluid forming a thin impermeable barrier that mitigate pressure transmission to the formation, combined with an eco-friendly material capable of sealing fractures up to 1,000 microns. Six years of field data of hundreds of field applications analyzed to assess top of cement (TOC) achievement and treatment reliability. Particular attention given to failed TOC cases to identify patterns and well conditions that affected outcomes. This structured approach aims to validate the treatment's overall performance and adaptability under varied operational scenarios.

The analysis revealed an average efficiency of 93% over six years, with two years achieving a perfect 100% success rate. Cases with failed TOC highlighted specific conditions, such as extreme fracture widths, presence of cavernous formations, or unusual well parameters, where the treatment was less effective. Despite these challenges, the lost circulation spacer demonstrated consistent reliability when paired with eco-friendly loss circulation materials, making it a robust solution in most scenarios. For wells with unique conditions leading to treatment failure, alternative loss circulation methods recommended. The study provides a comprehensive evaluation of the spacer's performance; field case examples and identifies critical considerations for optimizing future applications.

This paper presents a replicable evaluation framework, enabling engineers to maintain high success rates with lost circulation treatments. By analyzing six years of field data, the study offers actionable insights for improving treatment planning and addressing unique operational challenges.

Introduction

Lost circulation during drilling remains one of the most persistent challenges in the oil and gas industry, particularly in the Middle East. When losses occur and cannot be effectively cured prior to primary cementing, whether just before or during the job, it can lead to zonal isolation failure. This, in turn, may result in costly remedial operations or, in severe cases, the loss of the well.

Over the years, various lost circulation (LC) mitigation techniques during cementing have proposed and implemented, producing mixed results. A common approach involves the use of lost circulation materials (LCMs) such as fiberglass, crushed mica, flaked collophane, polymers, and other additives (Nelson and Guillot, 2006). However, these materials often prove ineffective, either failing to form a sufficient seal or being washed out or displaced before cement is in place.

This paper presents a six-year performance review of primary cementing jobs executed in naturally fractured zones and weak formations, with a focus on evaluating the efficiency of a wellbore shielding spacer system combined with an environmentally friendly LCM. It highlights key lessons learned, identifies areas for improvement, and offers recommendations to mitigate the risk of fluid losses during cementing.

Technology Background

The LC spacer is a combination of long chain polymers and flexible solids particles of different shapes and sizes that acts against the pore throats of the formation creating an impermeable barrier avoiding the pressure transmission to the formation. The spacer solids particles are activated once in contact with the problematic formation that requires to be sealed by the overbalance pressure in the wellbore, creating a thin impermeable barrier, resistance to erosion and high differential pressure. Once the wellbore is in underbalance pressure during the testing period or production phase the solids particles are easily reverse by the fluid flow.

In Fig. 1 is shown a comparison between the field deployment of a conventional spacer (left) and the LC spacer (right), on the illustration of the left can be observed that as the heavier fluid column while cementing raise in the annulus fluid is lost through the formation pore throats, inducing damage to the formation and limiting the height of the fluid column, adding additional risk to achieve zonal isolation. On the left, the LC spacer creates a thin impermeable barrier avoiding fluid loss to the formation, enhancing fluid coverage in the annulus and minimizing formation damage.

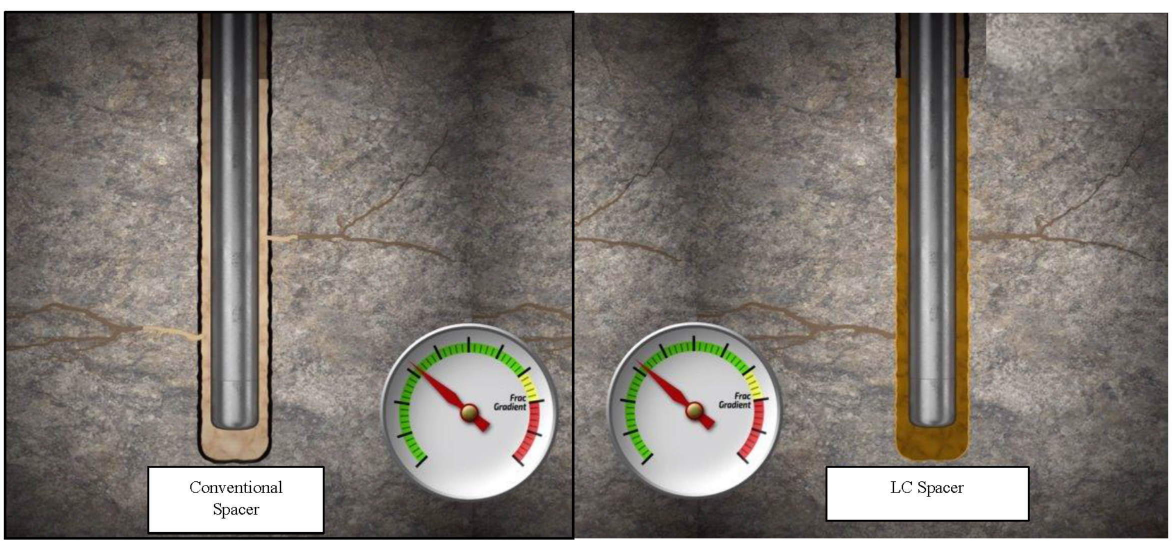

Figure 1—Illustration of a conventional spacer deployment in the wellbore showing potential losses to the formation (left) and the application of the LC spacer on the right creating an impermeable barrier, avoiding pressure transmission, enhancing fluid coverage and mitigating formation damage.

The LC spacer shows superior compatibility with various drilling fluids, including aqueous and non-aqueous systems and with cement slurries, maintaining the rheological properties of the mixture fluids within the friction pressure hierarchy and avoiding higher friction pressures during the cement operation. It is also compatible with most common surfactant's packages and solvents required to water-wet the annulus surfaces required for cement bonding.

Its typical concentration is 15 lb/gal, as it is composed of a mixture of polymers and solids particles that provides the shielding component, it's rheology is easy adjustable by modifying the concentration of the spacer mix, and can be used in densities ranging from 8.7 lb/gal to above 19 lb/gal. As a rule of thumb, higher the density, lower is the required concentration to achieve the desired rheological profile. In Fig. 2 is represented different concentrations at various densities of the LC spacer at surface (80 deg F) and at 170 deg. F. with the resulting YP, a liner behavior can be observed at both temperatures.

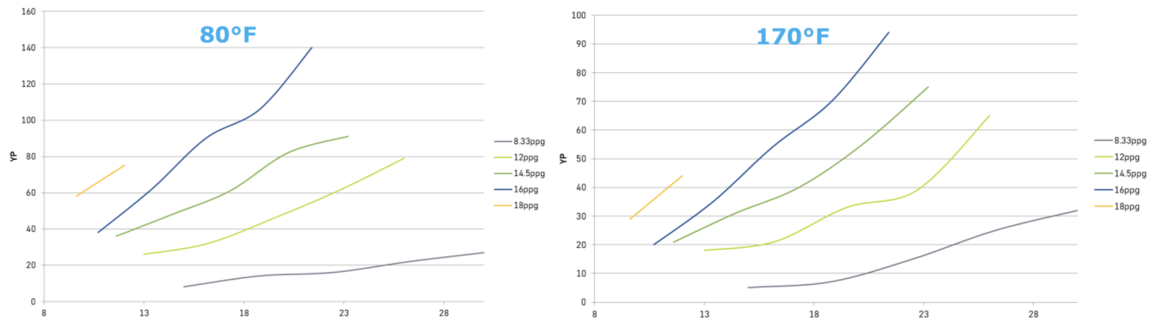

Figure 2—Common required concentration of the LC spacer at different densities and product loading and the expected Yield Point (YP). On the left resulting YP at surface and in the left at 170 deg F.

Benefits

- The wellbore shielding spacer provide enhanced mud removal, creating an interphase between the drilling fluids and the cement. The solids particles of the wellbore shielding spacer support the removal of the gelled mud from both the casing and the formation.
- Has a plugging capability up to 250 microns and by pairing with a natural occurring environmentally friendly loss circulation material its plugging capability can be extended to 1,000 microns or 3500 Darcies.
- Stops pre-existing losses if observed prior cementing and prevents induced loss circulation during cementing.
- Allows cement circulation to the programmed top of cement, preventing cement fall back.
- Avoids the cement slurry filtrate loss to the formation enabling proper setting and development of the design mechanical properties.
- Allows higher equivalent circulating density while cementing, mitigating formation breakdown improving zonal isolation and wellbore integrity.
- The LC spacer shows suitable rheological properties of up to 350 deg. F (175 deg C).

System Validation

The LC spacer prior field applications undergo all the standard laboratory testing as per API RP 10B-2, including:

- Rheological properties to determine its surface and downhole plastic viscosity (PV) and Yield point (YP).
- Fluids compatibilities with field mud sample and the lead slurry system, mandatory for oil-based mud (OBM) / synthetic-based mud (SBM) systems.
- BP Settling test or Static stability test.

- Wettability test or Surfactant Screening Test (SST), required for non-aqueous drilling fluids systems.

In addition to the above-mentioned tests, the LC spacer is tested to ensure its plugging capability and thermal stability. A Particle Plugging Apparatus (PPA) is used to test the plugging capacity and efficiency of the LCM spacer, as shown in Fig. 3. The LC spacer mixture is subjected a 1,000-psi differential pressure from top to bottom. At the boom of the PPA cell a slot test is placed, for the LC spacer a 250 microns slot is used and when paired with the LCM, a 1,000 microns slot. The initial fluid spurt and the total fluid loss after 15 min. are collected and weighted. In Fig. 3 is shown the PPA and the used slots can be observed prior and after the test, showing the formed filter cake.

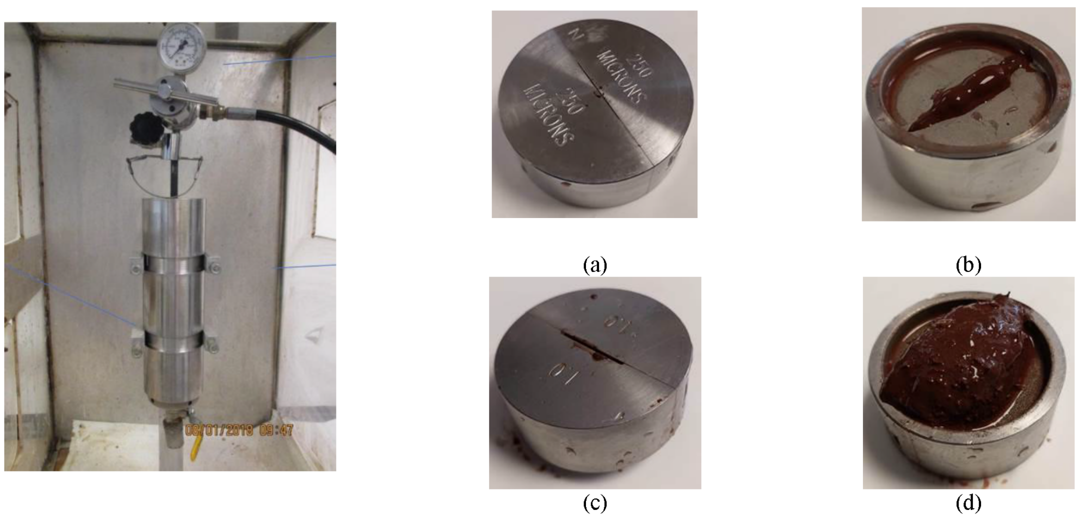

Figure 3—PPA and the 250 microns slot prior; (a) and after testing (b) the LC spacer. The 1,000 microns slot used to test the LC spacer paired with environmentally friendly LCM prior (c) and after (d) the test.

To test the thermal stability a lab sample of the LC spacer is prepared and poured into the high-pressure high-temperature (HPHT) consistometer and exposed to the field application downhole conditions. If it is expected that the fluid will remain static under downhole conditions, a GO-NO-GO test is included in the schedule by switching off the motor for the expected time the fluid may remain static, example of this kind of operations are liner cementing operations, where contingencies includes a potential static time of 45 min up to 1 hr., example shown in Fig. 4, where no degradation of the fluid is observed under downhole conditions.

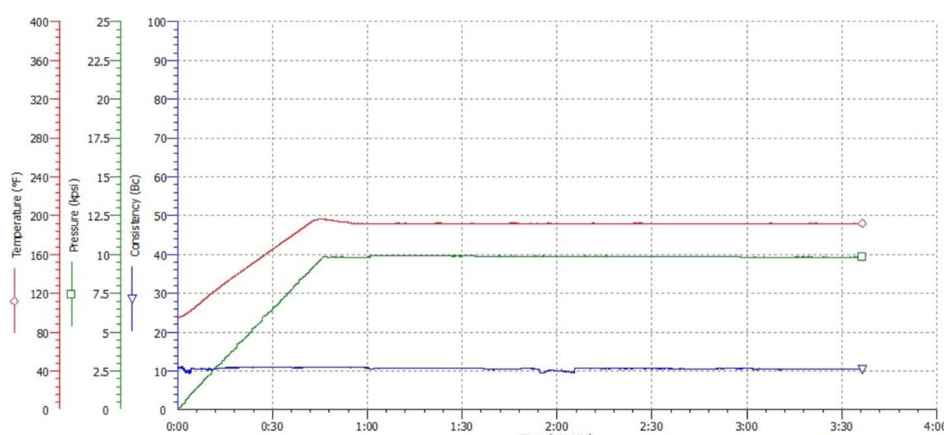

Figure 4—Example of a dynamic stability test performed to the LC spacer, under downhole conditions

Once the dynamic test is completed the slurry cup and paddles are visually inspected for any sign of solids sagging or accumulation on the paddle. In Fig 5. Can be observed that there is no accumulation of solids in the paddle, or inside and the bottom cap of the slurry cup.

Figure 5—Visual inspection of the paddle (left), cup (middle) and bottom cap (right) of the slurry cup after completion of the dynamic stability test

The same LC spacer sample exposed to downhole conditions is then poured into a 250 ml graduated cylinder to perform the static stability test for 2 hrs. After completion of the test the cylinder is inspected for any sign of free water and with a syringe, fluid sample from top, middle and bottom is taken to measure any potential density variation in the LC spacer, as shown in Fig. 6.

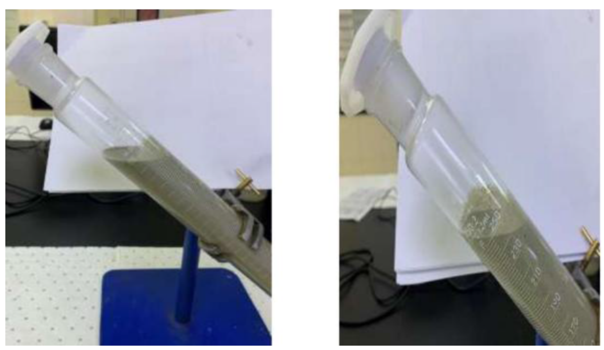

Figure 6—Static free fluid test on the left initial fluid sample. On the right the LC fluid sample after 2 hrs.

Methodology

This study is based on a six-year field performance review (2019-2024) of 188 cementing jobs executed in naturally fractured zones and mechanically weak formations across various casing and liner sizes. The focus was to evaluate the effectiveness of a wellbore shielding spacer system combined with an environmentally friendly LCM in minimizing cement losses and ensuring successful achievement of the TOC.

The dataset categorized by casing type, job year, and TOC achievement. A variety of cementing job types, spanning surface, intermediate, and production sections, by incorporating both conventional and specialized applications such as three-stage jobs, managed pressure cementing (MPC), and remedial operation such as LC cement plugs.

TOC achievement is used as the primary indicator of success. Trends assessed by comparing yearly TOC success rates across casing types. The job outcomes analyzed to:

- Quantify TOC success rate for each casing size.
- Evaluate performance improvements over time.

- Identify job types or conditions correlated with TOC failure.
- Highlight the effectiveness of the wellbore shielding spacer in high-risk intervals.

Statistical review and comparative analysis enabled the identification of persistent challenges and areas for operational adjustments, particularly in liner jobs and large-diameter casings where natural fractures and pressure-depleted zones are more prevalent.

Results and Discussion

Overall Efficiency

A total of 188 primary cementing operations were analyzed between 2019 and 2024. The success of each job is measured by whether the TOC was achieved. Table 1 provides a yearly breakdown of the number of jobs, TOC achievement, and corresponding success efficiency.

Table 1—Yearly cementing job efficiency summary

Year	Total jobs	TOC achieved	Failed TOC	Efficiency (%)
2019	16	13	3	81%
2020	24	22	2	92%
2021	19	19	0	100%
2022	44	41	3	93%
2023	43	40	3	93%
2024	42	42	0	100%
Total	**188**	**177**	**11**	**Avg: 93%**

The highest efficiencies observed in 2021 and 2024 (100%), demonstrating significant process maturity over time. Earlier years, no tably 2019, reflected challenges in managing TOC due to less refined job design and loss control practices.

Importantly, none of the 188 jobs analyzed involved the use of any LCM blended directly into the cement slurry (e.g., cement fibers), emphasizing the standalone effectiveness of the wellbore shielding spacer system.

Fig. 7 provides a visual summary of the cementing activity and TOC performance:

- Fig. 7a: Total cementing jobs analyzed per year from 2019 to 2024
- Fig. 7b: Cumulative TOC success versus failure cases across the dataset
- Fig. 7c: TOC results segmented by cementing job type (Primary cementing casing, Primary cementing Liners and LC Plugs)

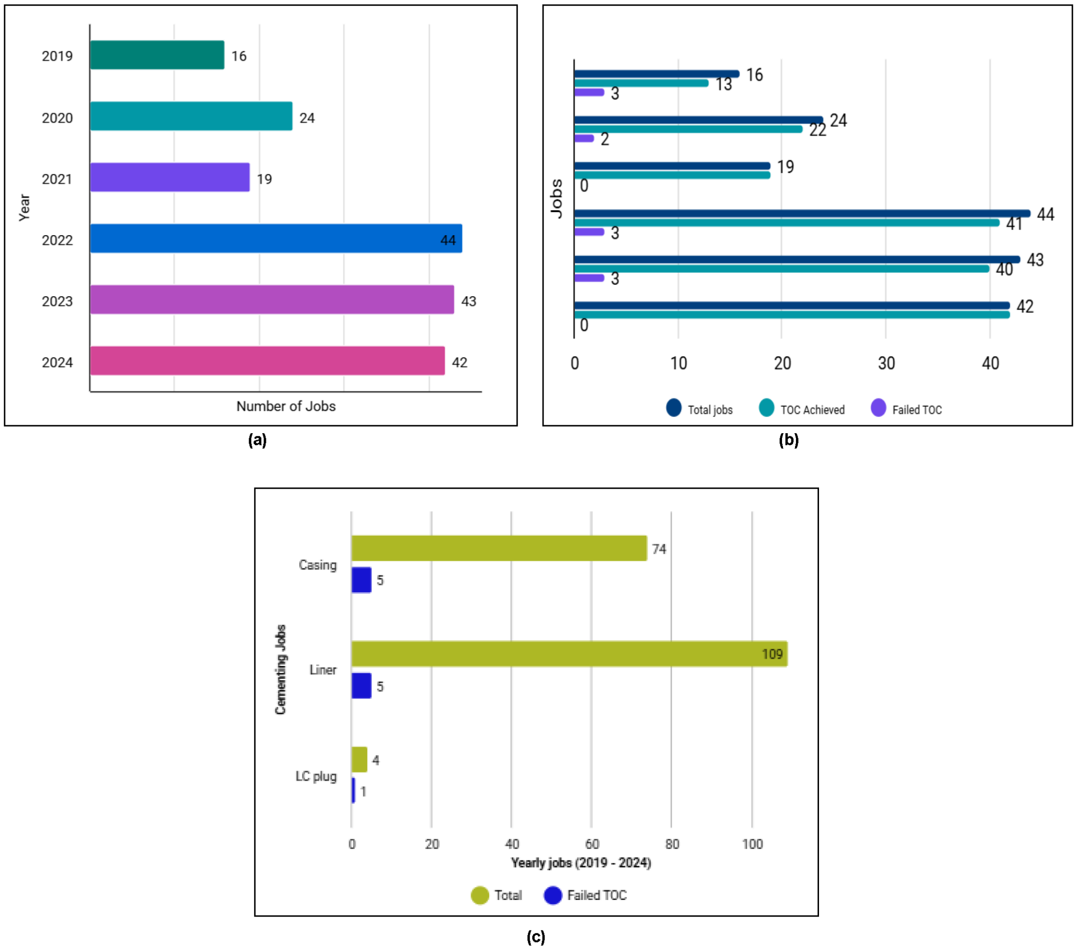

Figure 7—Visualization of Cementing Job Data and TOC Performance. (a) Cementing jobs from 2019 to 2024. (b) Total jobs versus the achieved TOC and failed TOC. (c) TOC achieved and failed versus cementing jobs

Analysis of Failed Cases

A closer review of the 11 failed TOC cases reveals important operational patterns:

Loss Conditions. All failed cases occurred under total losses or loss rates exceeding 200 barrels per hour (BPH). Out of 18 total losses jobs executed during the study period, only 7 succeeded in achieving TOC, representing a 39% efficiency for this subgroup, which are presented in the Fig. 8.

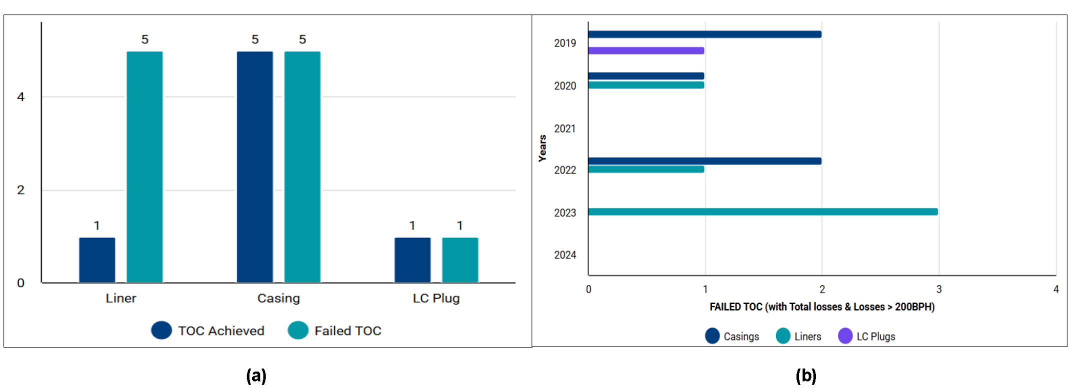

Figure 8—It depicts the cement job categorized with failed TOC from 2019 to 2024. (a) Part 1 TOC achieved or failed with cement jobs. (b) Part 2 Failed TOC with losses

Unknown Well Geometry & Washouts. Several failed jobs affected by uncertain wellbore geometries, likely exacerbated by formation washouts in highly fractured zones. The excess annular volume may have made cement volumes insufficient, leading to TOC failure.

Contributing Operational and Desing Factors.

- In many cases, the exact depth and extent of the loss zones were not well-defined before the job. This was particularly critical in formations with multiple or layered fractures, where losses could initiate at various intervals, complicating the cement design and volume estimation.
- Cement volumes and fluid systems not adequately adjusted to account for potential washout in the hole. In high-loss zones, underestimating the actual annular volume can lead to early termination of displacement, resulting in deeper TOC or fallback.
- Lack of real-time downhole data to monitor effective displacement or bottomhole pressure.

Lessons Learned

Key success factors.

Consistent TOC Success in Low to Moderate Losses

The implementation of the wellbore shielding spacer system, combined with an environmentally friendly LCM, consistently achieved the required TOC in wells experiencing low to moderate losses. The treatment's sealing and bridging capability proved effective in mitigating minor fluid migration during placement.

Robust Performance Near Fracture Gradient

The system demonstrated reliable results in scenarios where the Equivalent Circulating Density (ECD) during cementing approaches or slightly exceeded the formation fracture pressure. This highlights its utility in operations with narrow pressure margins, where conventional cementing systems risk fracturing the formation and losing cement slurry.

Outstanding Results in MPC Applications

The technology was particularly successful in liner cementing using the MPC technique. Across 17 liner jobs executed with MPC in complex wells, characterized by weak formations intersecting high-pressure zones, the shielding spacer contributed to 100% TOC achievement. These results align with previous findings (Masoud et al., 2024; Quintero et al., 2024), affirming its critical role in zonal isolation under such challenging conditions.

No Reported of sustained casing pressure(SCP) Cases

From all 188 cementing jobs where the wellbore shielding spacer was used, no SCP was reported at the time of publication. This strongly supports the system's long-term effectiveness in delivering reliable zonal isolation.

Improvement opportunities.

Alternative LC treatment for total loses scenarios

In cases of total or severe fluid losses (defined as losses exceeding 200 BPH), the wellbore shielding spacer system alone did not prove effective as a lost circulation solution. During the initial two years of the study, two LC cement plugs executed using this system, with only one achieving the desired TOC, resulting in a 50% success rate. Based on this performance, alternative LC treatments, such as engineered cement plugs or specialized loss-prevention fluids were recommended from 2021 onward to address high-loss conditions more reliably.

Development of high-performance LCM

There is an opportunity to develop or introduce an enhanced LCM that can be integrated into the wellbore shielding spacer system. This additive should be capable of sealing larger fractures (e.g., up to 3-5 mm aperture) while maintaining compatibility with downhole equipment, such as float collars, centralizers, and liner hanger restrictions. The goal is to enhance sealing performance in severe loss zones without compromising tool functionality or placement efficiency.

Job design adaptations must account for.

Real-time loss severity classification

Understanding the type of losses and depth is essential for the effectiveness of the treatment.

Potential washout volume adjustments

Increasing the annular excess to account for the washout volume uncertainties.

LCM type and concentration

More aggressive LCM blends or bridging agents can be considered for severe losses treatment. In case of inability, it is possible to increase the concentration of the current LCM.

ECD management

Managing the ECD to avoid further losses across the already fractured formation nor another weaker formation in the section.

Correlations and Takeaways.

Loss Type vs. TOC Success

A strong correlation observed between moderate loss scenarios (losses <200 BPH) and successful TOC outcomes. This reinforces the value of the shielding spacer system as a proactive loss mitigation solution, particularly in naturally fractured or weak formations where partial losses anticipated.

Effectiveness in MPC

The wellbore shielding spacer system demonstrated critical value in MPC operations, where maintaining zonal isolation is paramount. Across 17 MPC liner jobs, 100% TOC success recorded. This is especially relevant in wells that intersect weak zones and high-pressure formations, where failure to isolate zones could pose serious safety, environmental, and production risks.

Strategic Approach to Loss Management

The success of any loss mitigation method, including the use of wellbore shielding spacer, relies heavily on effective pre-job diagnostics, offset well data, and a structured loss circulation strategy. Persistent or severe losses must be evaluated and, if necessary, treated prior to the cementing operation to maximize the probability of achieving zonal isolation. A predefined LC treatment matrix should be part of the drilling program, enabling selection of the most suitable solution depending on the severity, type, and depth of the losses.

Case Study Highlights

Case study #1

A production gas well experienced partial losses after drilling the 5 7/8" open hole (OH) section, and after running in hole (RIH) the 4 ½" liner at total depth (TD). The loss rate was recorded at 96 BPH. To address the losses, a 50 bbl pill of wellbore shielding spacer, mixed with 25 lb/bbl of an environmentally friendly LCM, was pumped prior to the cementing operation. This initial treatment effectively reduced the losses to 36 BPH.

Based on this improvement, the primary cementing operation proceeded with 75 bbl of the same shielding spacer system, followed by a 125 pcf cement slurry. After bumping the plug and setting the liner top packer, reverse circulation was established above the top of liner (TOL), successfully recovering both spacer and pure cement slurry. Subsequent testing confirmed effective zonal isolation across the section, and the well was completed and handed over for production.

Post-execution, the cementing data was analyzed using cementing software simulation. The pressure match analysis in Fig. 9 revealed a close alignment between the actual pressure (blue line) and the simulated pressure (red line) throughout the job. No indication of losses was observed, suggesting that the shielding spacer combined with the eco-friendly LCM effectively created an impermeable film in the formation, allowing the cement slurry to reach the top of the liner.

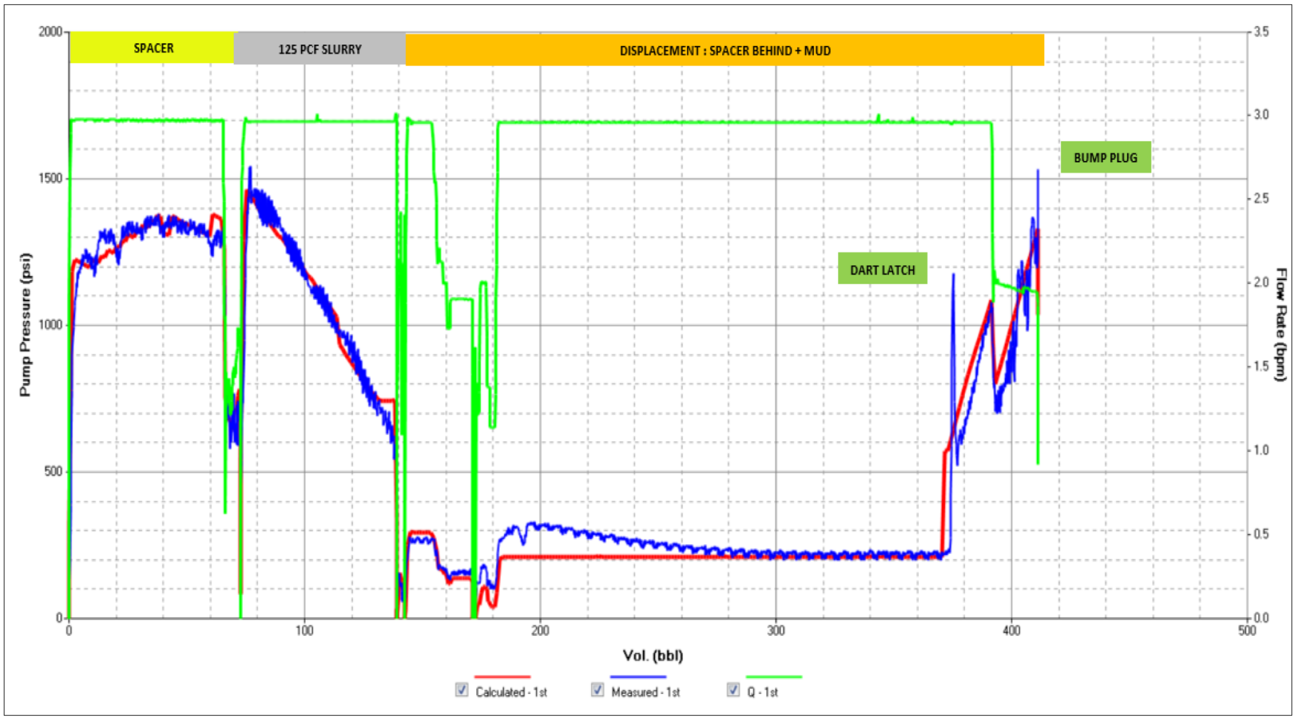

Figure 9—Pressure match analysis Case Study #1.

The application of the shielding spacer system allowed the operator to avoid pulling out the liner to address the losses—a scenario that would have resulted in non-productive time (NPT) and increased operational costs.

Case study #2

The 8 3/8" hole section of a wildcat well was drilled with 71 pcf OBM under full circulation. However, the maximum ECD while cementing the 7" liner was expected to exceed the fracture gradient by at least 3 pcf. A specific hydrocarbon target zone in the section required complete isolation from adjacent water-bearing formations, and the risk of losses posed a challenge to achieving this.

To mitigate the risk, 80 bbl of 95 pcf wellbore shielding spacer was pumped, followed by 118 pcf lead slurry and 127 pcf tail slurry. The cementing operation was executed successfully, dart latched was observed, and the liner wiper plug bumped at 3200 psi. After setting the liner top packer, reverse circulation was established above the TOL, returning both spacer and cement to surface. Cement evaluation logs confirmed that the TOC was achieved, and the target zone was fully isolated.

As shown in Fig. 10, the actual and simulated pressures closely matched, indicating that no losses occurred during the operation.

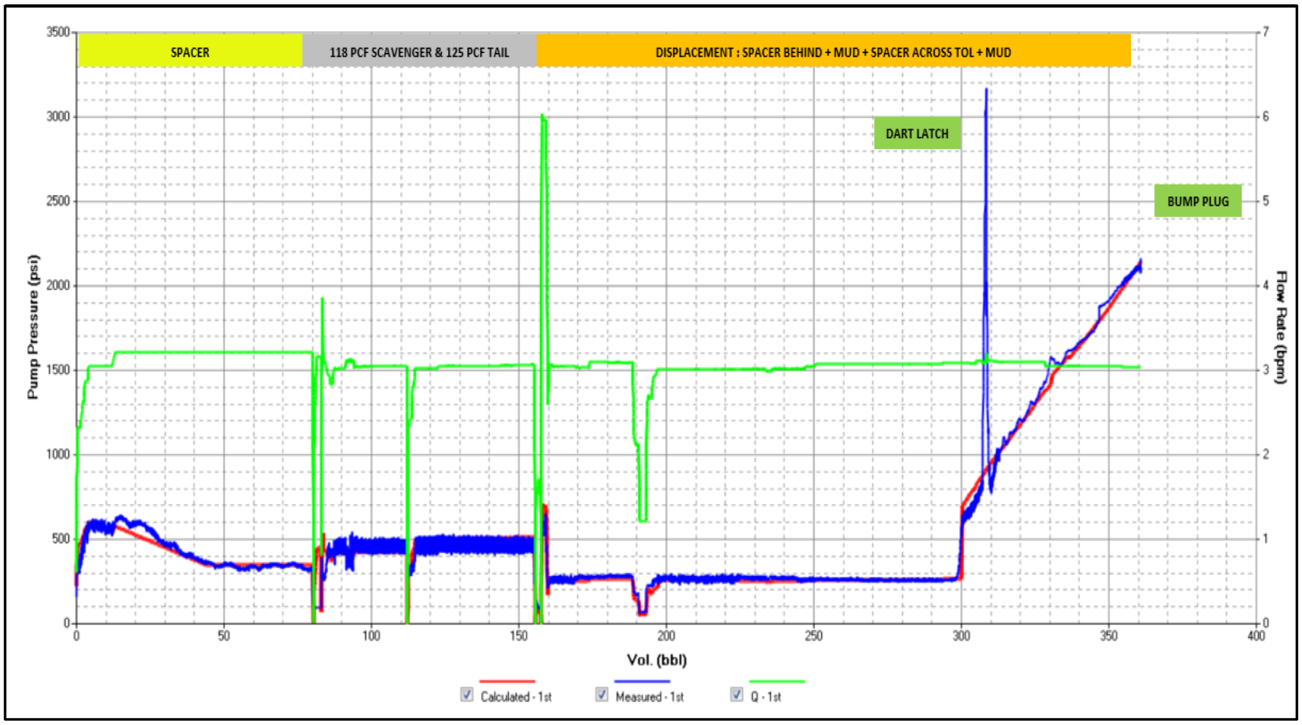

Figure 10—Pressure match analysis Case Study #2.

Case study #3

The 5 7/8" OH section was drilled using 95 pcf OBM with full circulation to TD. After RIH the 4 ½" liner and setting the liner hanger, the well encountered dynamic losses; 18 BPH at 4 barrel per minute (BPM) and 6 BPH at 3.5 BPM. To address this, 80 bbl of shielding spacer was pumped ahead of cement, followed by 112 pcf lead and 118 pcf tail slurries. The cement was displaced with spacer and mud at a constant rate of 3.5 BPM.

Dart latch and wiper plug bump were clearly observed in the pressure chart. The plug was bumped with 1,000 psi over the final displacement pressure. Following the liner top packer setting, reverse circulation was established, with cement returned to surface, achieving TOC across the targeted zones.

After waiting on cement (WOC) period, the rig successfully performed both positive and negative pressure tests, validating zonal isolation.

Post-job analysis in Fig. 11 revealed close alignment between actual and simulated pressures, with no anomalies observed. This confirmed the shielding spacer's effectiveness in preventing cement slurry losses and achieving TOC.

Figure 11—Pressure match analysis Case Study #3

Case study #4

A re-entry well was drilled from a window opened in the 9 5/8" casing, and a deviated 8 3/8" section was drilled using managed pressure drilling (MPD). Based on pore pressure and losses recorded, the operational window was determined to be just 1 pcf wide, with fluid influx occurring below 143 pcf ECD and losses occurring above 144 pcf.

While drilling with 129 pcf OBM and maintaining 144 pcf equivalent mud weight (EMW) using MPD, the well experienced constant partial losses (~30 BPH). The 7" liner was cemented using the MPC technique, with 170 bbl of 138 pcf wellbore shielding spacer paired with eco-friendly LCM, followed by 150 pcf cement slurry.

Following displacement, plug bump, and liner top packer setting, reverse circulation returned both spacer and cement slurry, indicating TOC was reached. After WOC, liner pressure tests and negative tests confirmed isolation.

The pressure match analysis in Fig. 12 shows no indication of losses, affirming the effectiveness of the shielding system in this narrow-margin well.

These four case studies demonstrate the versatility and effectiveness of the wellbore shielding spacer system in preventing and mitigating cement slurry losses, without the need of any additional LCM added to the cement slurry, enabling successful TOC and zonal isolation even in complex and high-risk environments, preventing expensive and complicated remedial treatment.

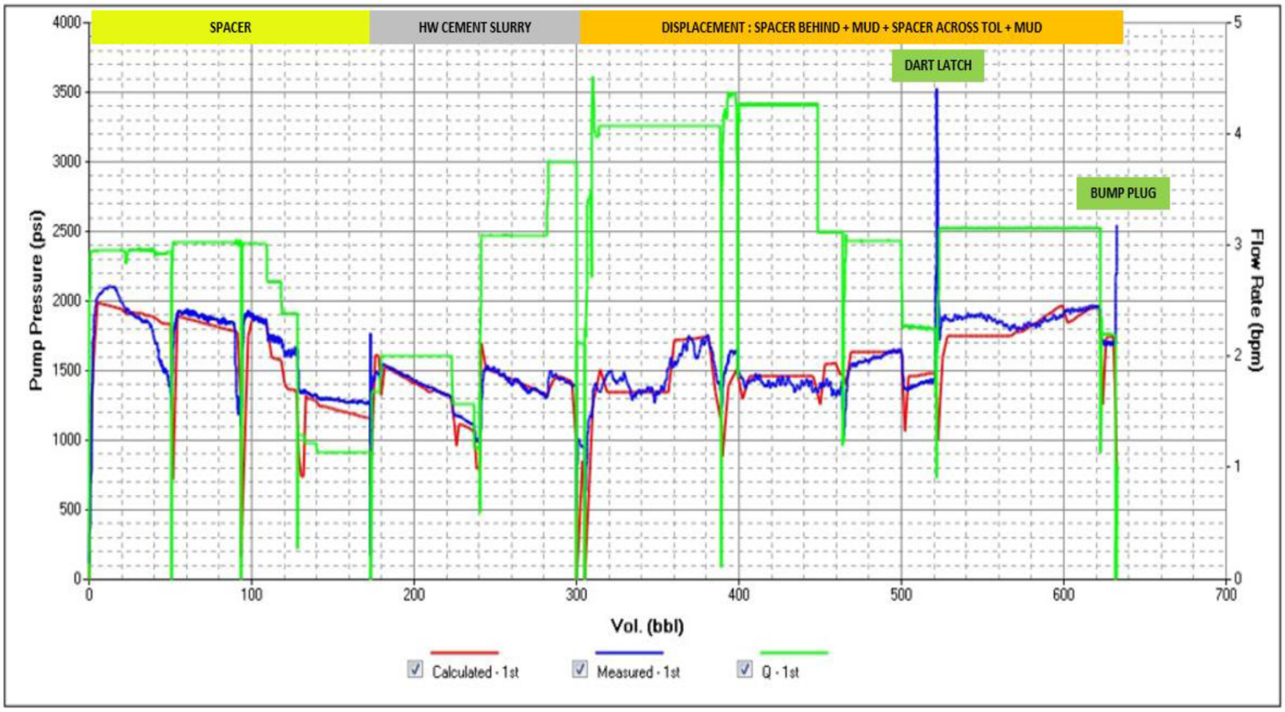

Figure 12—Pressure match analysis Case Study #4

Conclusion

The six-year performance review of 188 primary cementing operations in loss-prone formations demonstrated that the use of a wellbore shielding spacer system, paired with an environmentally friendly LCM, significantly improved cementing success in moderate to partial loss conditions. With an overall TOC success rate of 93%, and 100% success in MPC jobs, the system has proven to be a reliable tool for achieving zonal isolation and reducing the need for remedial operations.

However, the system showed limited effectiveness in total loss and severe loss scenarios (losses >200 BPH), where unpredictable well geometry and uncontrolled fluid losses reduced the likelihood of success. These outcomes underline the importance of accurate pre-job diagnostics, early identification of potential loss zones, and the use of alternative or supplementary loss circulation strategies when needed.

The results also reinforce the value of pairing mechanical and fluid-based loss mitigation techniques, particularly in complex well environments. Notably, no SCP was recorded in any of the treated sections at the time of this publication, supporting the long-term reliability of the approach.

Going forward, the continued development of high-performance LCMs compatible with wellbore shielding spacer systems, along with more robust pre-cementing evaluation workflows, will be key to expanding the applicability and efficiency of this technology, especially in more challenging loss scenarios.

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Nomenclature

API =*American Petroleum Institute*
BPH =*Barrels per hour*

BPM =*Barrel per minute*
ECD =*Equivalent Circulating Density*
EMW =*Equivalent Mud Weight*
HPHT =*High-Pressure, High-Temperature*
LC =*Lost Circulation*
LCM =*Lost Circulation Material*
MPC =*Managed Pressure Cementing*
MPD =*Managed Pressure Drilling*
NPT =*Non-productive time*
OH =*Open Hole*
OBM =*Oil-Based Mud*
PCF =*pounds per cubic foot*
PPA =*Particle plugging apparatus*
PSI =*pounds per square inch*
PV =*Plastic viscosity*
RIH =*Run in Hole*
SBM =*Synthetic-Based Mud*
SCP =*Sustained casing Pressure*
SST =*Surfactant Screening Test*
TD =*Total Depth*
TOC =*Top of Cement*
TOL =*Top of Liner*
WOC =*Waiting on Cement*
YP =*Yield point*

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